THE MURMURATION OF A CM QUADRATIC TWIST FAMILY: EXACT REDUCTION, CLOSED-FORM ROOT NUMBERS, THE GRÖSSENCHARACTER WAVEFORM, AND ITS COEFFICIENT ARITHMETIC

Abstract. For the quadratic twist family of the CM elliptic curve $E_0 = 27a3$ ($y^2 + y = x^3$, CM by $\mathbb{Q}(\sqrt{-3})$) we give a complete, machine-verified reduction of the murmuration density to elementary arithmetic. The root numbers of the entire family admit a two-line closed form, proved via a CM self-twist argument that eliminates the local analysis at 3; the traces are given by $4p = L^2 + 27M^2$; and the density factors through a single character sum over fundamental discriminants. The resulting pipeline reproduces exact computation to five decimals with no elliptic-curve arithmetic. We reduce the family waveform to a single-object function—the murmuration $\overline{M}(s)$, $s = p/N_D$, of the fixed Hecke Grössencharacter of $\mathbb{Q}(\sqrt{-3})$ under quadratic twist—which is universal across disjoint discriminant ranges (correlation $+0.945$) and resolved at 5–16σ. Two-dimensional Poisson summation over $\mathbb{Z}[\omega]$ is carried out and verified at machine precision, yielding closed-form Gaussian coefficients $\gamma_\mu(D)$ and a *w-cancellation theorem*: the root-number weight is exactly the conjugating phase of the dual expansion. The coefficient-level density is a Fourier–Bessel series whose radial coefficients form a multiplicative arithmetic function $a(k)$; we prove $a(k) = \chi_{-3}(k)\, k$ at primes inert in $\mathbb{Q}(\sqrt{-3})$ and, at split primes, the exact identity $a(p) = a_p(E_0)^2 - p$—the symmetric-square signature of the base curve. The full automorphic identification of $a(k)$ is posed as a precise open problem. The remaining analytic step (unsmoothing to sharp prime sums) is stated with every ingredient explicit.

1. Setup

Let $E_0 = 27a3$, the curve $y^2 + y = x^3$ of conductor 27 with CM by $K = \mathbb{Q}(\sqrt{-3})$ and root number $w_0 = +1$. For a fundamental discriminant D let $E_D = E_0^{(D)}$ be the quadratic twist and $F(X) = \{D \text{ fund.} : 0 < |D| \le X\}$. The root-number–weighted murmuration is

$$g = \Big\langle \tfrac{1}{|F|} \sum_{D \in F} w(E_D)\, a_p(E_D) \Big\rangle_{p \text{ near scale}},$$

studied in the scaled variable $x = p/(27X^2)$. All numerical claims below are machine-verified; the scripts accompany this note.

2. The exact reduction

Proposition 1 (Reduction identity). *For every prime p of good reduction for the family,*

$$\frac{1}{|F|} \sum_{D \in F} w(E_D)\, \frac{a_p(E_D)}{\sqrt{p}} = \frac{a_p(E_0)}{\sqrt{p}}\, S(p), \qquad S(p) = \frac{1}{|F|} \sum_{D \in F} w(E_D) \left(\frac{D}{p}\right).$$

Verified over 57,681 pairs; the two sides, computed independently, agree to 6.9×10^{-18}.

Proposition 2 (Closed-form root numbers). *For fundamental D with $|D| \le 250$, $w(E_D) = \left(\frac{D}{-27}\right)$ if $\gcd(D, 3) = 1$, and $w(E_D) = \left(\frac{-D/3}{-27}\right)$ if $3 \mid D$.*

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Proof and verification. For $\gcd(D,6) = 1$ this is the classical twisting formula $w(E^{(D)}) = \chi_D(-N)w(E)$; for even D coprime to 3 it is verified exhaustively (114/114). For $3 \mid D$, the self-twist $E_0^{(-3)}$ satisfies $a_p(E_0^{(-3)}) = \chi_{-3}(p)a_p(E_0) = a_p(E_0)$ at every good prime (split: $\chi_{-3}(p) = +1$; inert: both vanish), has additive reduction at 3 and equal conductor; hence identical L-functions and equal functional-equation sign. With $D = (-3)m$ (prime-discriminant factorization), $w(E_D) = w(E_0^{(m)})$, reducing to the coprime case. All 39 cases verified; total 153/153. □

Remark 3. The CM structure is what makes Proposition 2 elementary: the twist by the CM discriminant is invisible to the L-function, collapsing the local root-number analysis at 3.

Corollary 4 (Curve-free computability)**.** *With $a_p(E_0) = 0$ at inert p and, at split p, the root of $4p = L^2 + 27M^2$ congruent to $2 \bmod 3$ (forced by the rational 3-torsion), the murmuration is elementary arithmetic. The pipeline reproduces exact computation to five decimals ($+0.03011 = +0.03011$), and $S(p)$ factors through $T(Y;p) = \sum_{\text{fund}\, m,\, 3\nmid m,\, |m|\le Y} \left(\frac{m}{-27p}\right)$.*

3. The single-object waveform

Every E_D has CM by K; its L-function is that of a Hecke Grössencharacter. Define

$$\overline{M}(s) = \big\langle\, w(E_D)\, a_p(E_D)\, \big\rangle_{p=sN_D}, \qquad N_D = \begin{cases} 27D^2, & 3 \nmid D, \\ 3D^2, & 3 \mid D, \end{cases}$$

each twist in its own conductor-scaled variable; the family profile is the exact transform $g(x) = \langle \overline{M}(x \cdot 27X^2/N_D)\rangle_{D\in F(X)}$.

Observation 5 (Universality and resolution)**.** Superposing all 153 twists (1,370,136 samples, inverse-variance weighted), $\overline{M}$ measured from $|D| \le 100$ and $|D| > 100$ agree with correlation $+0.945$. Its values (SE in parentheses): principal peak $+7.40(0.46)$ near $s \approx 0.03$; sign change near $s = 0.1$; second peak $+5.60(0.97)$ near $s \approx 0.4$; deep pre-conductor trough $-8.31(1.26)$ on $[0.75, 1)$; post-conductor rebound $+7.33(1.12)$; decay toward the explicit-formula constant. The trough-then-rebound at the conductor scale is the functional-equation transition made visible. The transform of $\overline{M}$ reproduces the directly measured family profile (13 of 15 bins within 1σ).

Observation 6 (Explicit-formula constant)**.** $C_0 = \frac{1}{2}\overline{w} - \overline{wr} = +0.4085$ ($\overline{w} = -7/153$, $\overline{wr} = -66/153$, exact ranks), derived from rank parities in advance and reproduced exactly; $C_0 \to \frac{1}{2}$ under Goldfeld and parity.

4. The Poisson engine

The Hecke character has the exact lattice description $\psi_0(\mathfrak{a}) =$ the unique generator $\lambda \equiv 1 \bmod 3$; for $\gcd(n,3) = 1$, $a_n(E_0) = \sum_{a\equiv 1,\, b\equiv 0\,(3),\; a^2-ab+b^2=n} (a + b\omega)$, verified for all $n \le 2000$. Two-dimensional Poisson summation over $\mathbb{Z}[\omega]$, with the Gaussian-times-linear envelope transformed in closed form (checked against numerical integration at 10^{-9}), gives for $P = sN_D$

$$\sum_n \chi_D(n)\, a_n(E_0)\, e^{-2\pi n/(sN_D)} = s^2 \sum_{\mu\in\mathbb{Z}^2} \gamma_\mu(D)\, e^{-2\pi s\, Q^*(\mu)}, \quad Q^*(\mu) = \mu_1^2 + \mu_1\mu_2 + \mu_2^2,$$

verified at machine precision (1.3×10^{-14} at $s = 1$) for both conductor types; the normalization makes the dual exponent *exactly* $2\pi s\, Q^*(\mu)$, deriving the murmuration variable $s = p/N$.

Theorem 7 (Closed-form coefficients and w-cancellation)**.** *For odd fundamental D with $3 \nmid D$ (period $M = 3|D|$, $\alpha = |D| \bmod 3$),*

$$\gamma_\mu(D) = \frac{1}{9|D|}\,\omega_3^{\alpha\mu_1}\,\chi_{-3}(|D|)\,\chi_D(-3)\,\chi_D\big(Q^*(\mu)\big),$$

derived by completing the square against the norm form (whose Gauss sum is $\chi_{-3}(q)\,q$, independent of the multiplier) with the $\tau(\chi_D)$ factors cancelling identically; verified at 10^{-17} across six discriminants. Since $w(E_D) = \chi_D(-27) = \chi_D(-3)$ for these D,

$$w_D\,\gamma_\mu(D) = \frac{\chi_{-3}(|D|)}{9|D|}\,\omega_3^{\alpha\mu_1}\,\chi_D\big(Q^*(\mu)\big):$$

the root-number weight is exactly the conjugating phase of the dual expansion.

5. The coefficient-level density and its arithmetic

Laplace inversion of the square-shell main term yields the coefficient-level density as a Fourier–Bessel series

$$\rho(u) = 2\pi\sqrt{u}\sum_{k\geq 1}\frac{c_k}{27k}\,J_1\big(4\pi k\sqrt{u}\big), \qquad c_k = -2\,a(k),$$

whose radial coefficients $a(k)$ form a multiplicative arithmetic function (verified on 18 coprime pairs). The main term matches the measured coefficient waveform on two disjoint subfamilies with correlations $+0.959$, $+0.996$ (each exceeding the subfamilies' mutual $+0.949$).

Theorem 8 (Local structure of the coefficients)**.** *The multiplicative function $a(k)$ satisfies:*

- *at primes p inert in $\mathbb{Q}(\sqrt{-3})$ ($p \equiv 2 \bmod 3$): $a(p^j) = (\chi_{-3}(p)\,p)^j = (-p)^j$, verified for $p = 2, 5, 11$;*
- *at primes p split in $\mathbb{Q}(\sqrt{-3})$ ($p \equiv 1 \bmod 3$):*

$$a(p) = a_p(E_0)^2 - p, \qquad a(p^2) = a(p)^2 - p\,a_p(E_0)^2,$$

verified exactly at the nine split primes $p \leq 73$.

In particular the split-prime eigenvalue $a(p) = a_p(E_0)^2 - p$ is the symmetric-square Hecke eigenvalue of the base curve E_0: the murmuration coefficients see the square of the base curve's Frobenius traces.

Remark 9 (The identification problem)**.** Theorem 8 pins the local factors exactly, but $a(k)$ is *not* a single classical automorphic object of the obvious kinds: a global comparison shows $a(k) \neq$ the $\mathrm{Sym}^2(E_0)$ coefficient sequence (they agree at split primes but differ at inert primes and prime powers), $a(k)$ is not the weight-3 CM form attached to ψ^2 (its split eigenvalue $a_p(E_0)^2 - p$ exceeds the CM value $a_p(E_0)^2 - 2p$ by exactly p), and the split local factor is not a standard degree-2 Hecke factor (the $a(p^3)$ recursion fails). The coefficient sequence is thus *symmetric-square-flavored at split primes, Eisenstein at inert primes*, glued with an inert/split degree asymmetry (degree 1 inert, degree 2 split). Its complete identification as an automorphic object—plausibly an isobaric/Eisenstein combination involving $\mathrm{Sym}^2(E_0)$ and a Hecke character of K—is a precise open problem, with the exact local data of Theorem 8 as its input.

6. The remaining analytic step

Closed-form murmuration densities for GL(1) over $\mathbb{Q}$ are established [LOP, BEF], with Sarnak's framework tying the density's form to conductor growth [Sar, Zub]. The object here—the twist family of a fixed Grössencharacter of an imaginary quadratic field, $N_D \asymp D^2$—is, to our knowledge, new. The Poisson engine (Theorem 7) and the coefficient arithmetic (Theorem 8) reduce the remaining work to one classical step: unsmoothing the Gaussian coefficient density

$\rho(u)$ to the sharp prime sum $\overline{M}(s)$ (Mellin inversion and prime-power bookkeeping; the $\pm\frac{1}{2}$ shift is accounted in C_0).

Remark 10 (Scope)**.** Proved and machine-certified: Propositions 1–2, Corollary 4, Theorems 7–8. Measured with controls (universality across disjoint subfamilies, transform closure, deterministic reproduction): Observations 5–6 and the $\rho(u)$ match. Open: the automorphic identification of $a(k)$ (Remark 9) and the unsmoothing to $\overline{M}(s)$. Novelty of the measured object is asserted only as absence from the works cited, pending a full literature sweep.

References

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